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Computer Architecture for Machine Learning - Section 2

AGResU-Net for Automatic MRI Brain Tumor Segmentation Enhancement & Results Report

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Subset Used:
https://drive.google.com/drive/folders/1zQEB96LPT6rXmfYbYmZFQs77UeWMlltg?usp=sharing

Overview

We introduce two enhancement techniques in phase 4, applied on top of the trained AGU-Net and AGResU-Net models: Test-Time Augmentation (TTA) and model ensembling. These techniques did not require any additional training and improved prediction quality by reducing model uncertainty.

Enhancement Techniques

➢ Test-Time Augmentation (TTA)

TTA runs each test slice through multiple augmented versions of the input, then averages the resulting probability maps before taking the final argmax. We used four augmentations:

- Original
- Left-right flip
- Up-down flip
- Both flips simultaneously

Each model makes predictions on all four versions independently. Then the flipped predictions are un-flipped before averaging to ensure spatial alignment. The final prediction is the mean of all four probability maps.

This method works because neural networks are not perfectly flip-invariant, they may segment a tumour more confidently when it appears in one orientation versus another. And by averaging across the orientations these inconsistencies will be smoothed out.

➢ Model Ensemble

After applying TTA to both AGU-Net and AGResU-Net independently, the two TTA probability maps are averaged to produce one single ensemble prediction. This uses the fact that the two models have different inductive biases: AGU-Net uses plain convolutional blocks while AGResU-Net uses residual blocks. Where one model is uncertain, the other may be more confident, and averaging reduces the overall error.

Impact of the Small Dataset

All experiments used a 160-patient subset of BraTS 2019 (the full dataset has 335 patients), split 60/20/20 into train/val/test. This had important consequences on our results:

- Lower base model performance: with less data, models generalise less well, especially for regions like whole tumour which require seeing many different slice appearances.
- Smaller gains from enhancement: TTA and ensembling work best when the base model is already strong. On a small dataset, the base model's predictions have higher variance, so the improvement ceiling is lower.

- Inverted DSC pattern: our models score higher on enhancing tumour (~0.78 - 0.80) than on whole tumour (~0.70 - 0.72). This is the opposite of the paper, because slice filtering retains only tumour-containing slices, which are disproportionately rich in enhancing tumour (label 3). The model sees more enhancing tumour examples relative to the diffuse edema that dominates whole-tumour scoring.

Despite these constraints, TTA and ensemble produced consistent gains, particularly on enhancing tumour.

Results

Model	DSC-Whole	DSC-Core	DSC-Enh	HD95-W	HD95-C	HD95-E
Paper Baselines († full BraTS 2019, 335 patients)						
U-Net†	0.864	0.746	0.696	-	-	-
ResU-Net†	0.867	0.760	0.704	-	-	-
AGU-Net†	0.870	0.760	0.700	-	-	-
AGResU-Net†	0.870	0.777	0.709	-	-	-
Our Results (160-patient subset)						
U-Net	0.701	0.712	0.768	9.23	5.42	3.14
ResU-Net	0.694	0.700	0.781	8.37	6.05	3.43
AGU-Net	0.711	0.702	0.783	9.98	6.51	3.10
AGResU-Net	0.721	0.694	0.770	10.60	5.68	3.50
AGU-Net + TTA	0.713	0.718	0.795	9.88	5.47	3.45
AGResU-Net + TTA	0.726	0.716	0.793	10.03	5.38	3.39
Ensemble (TTA)	**0.724**	**0.717**	**0.800**	**8.61**	**5.19**	**3.41**

† Paper values on full BraTS 2019 (335 patients). Our models trained on 160-patient subset.

Analysis

The table below shows the DSC improvement from the best individual base model to the Ensemble (TTA):

Region	Best Base	Ensemble	Gain
Whole Tumour	0.721 (AGResU-Net)	0.724	+0.003
Core Tumour	0.712 (U-Net)	0.717	+0.005
Enhancing	0.783 (AGU-Net)	0.800	**+0.017**

The enhancing tumour region saw the most meaningful gain (+0.017 DSC). This is significant as enhancing tumour is the most important region for treatment planning. Whole and core tumour gains were smaller but consistent.

For the HD95 case, it measures the 95th percentile surface distance between prediction and ground truth. The ensemble improved HD95 on the core tumour region (5.42 → 5.19) but unfortunately showed marginal degradation on whole and enhancing tumour compared to the single best individual model. This is expected behaviour where ensemble averaging softens sharp boundaries slightly, which can increase the maximum surface distance on some samples even while improving the overall DSC.